AI 321: Machine Learning for Sequential Data

System Models of Time-Series

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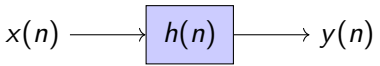
September 4, 2024

Overview

1. Analysis of DT-LTI Systems

2. Random Process

Analysis of DT LTI System



In general, the output $y(n)$ of a DT-LTI system with input $x(n)$ and impulse response $h(n)$ can be expressed as a linear combination of past outputs and passed inputs. That is

$$y(n) = \sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k)$$

The above difference equation is also called the **input-output equation** and represents one of the ways to describe a **discrete-time linear time-invariant** system (DT-LTI).

System Responses

The general solution of a difference equation be expressed as the sum of two functions.

$$y[n] = y_h[n] + y_p[n]$$

where $y_h[n]$ is the homogeneous solution (or complimentary function)
and $y_p[n]$ is the particular solution

System Responses

For the case that the difference equation represents a DT-LTI system, then the output $y[n]$ is expressed as:

$$y[n] = y_n[n] + y_f[n] = y_{zi}[n] + y_{zs}[n]$$

where $y_n[n]$ is the *natural response* and $y_f[n]$ is the *forced response*
also equal to the sum of the *zero-input response* the *zero-state response*.

(Note: In some cases, the natural response is equal to the zero-input response, and the forced response is equal to the zero-input response, but in general it is not always the case)

Analysis of DT-LTI systems

Decomposition: Decompose the input signal $x(n)$ as a sum of elementary signals so that the response $y(n)$ is easily determined, then apply linearity to obtain the total response. (example earlier)

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

Convolution: The output $y(n)$ can be computed by convolving the input $x(n)$ and the impulse response h(n).

$$y(n) = x(n) * h(n)$$

Analysis of DT-LTI systems

Iterative method: The output $y(n)$ is computed directly from the difference equation iteratively.

Direct method: Solving the closed-form solution of the of the input-output equation of the system.

Example

Consider the finite-duration sequence given as:

$$x(n) = \{2, \underset{\uparrow}{4}, 0, 3\}$$

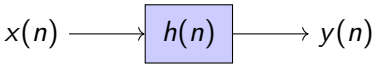
(Note: the $\uparrow$ indicates $n = 0$)

Resolving the sequence as a sum of impulses, we have:

$$x(n) = 2\delta(n+1) + 4\delta(n) + 3\delta(n-2)$$

Convolution Sum

- For an LTI system, we define $h(n)$ as the **unit impulse response**.



  if $x(n) \rightarrow \delta(n)$, then $y(n) \rightarrow h(n)$, that is $h(n) = H[\delta(n)]$
- The output response $y(n)$ is defined as:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

- the above equation is known as the **convolutional sum**
- we say the input is $x(n)$ is convolved with $h(n)$ to yield the output $y(n)$

$$y(n) = x(n) * h(n)$$

Convolution Sum

- The output of the system at an instant say $n = n_0$ is given as:

$$y(n_0) = \sum_{k=-\infty}^{\infty} x(k)h(n_0 - k)$$

That is

$$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(0 - k)$$

$$y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1 - k)$$

And so on.

- The index of summation is k, hence both the input signal $x(k)$ and impulse response $h(k)$ are functions of k.

Convolution Sum

- The output $y(n_0)$ is summed over all values of the product sequence $x(k)h(n_0 - k)$ for all values of k from $-\infty$ to $+\infty$.

- The signal $h(n_0 - k)$ is obtained by "folding" $h(k)$ about $k = 0$ $(h(-k))$, then shifting it by n_0 to obtain $h(n_0 - k)$

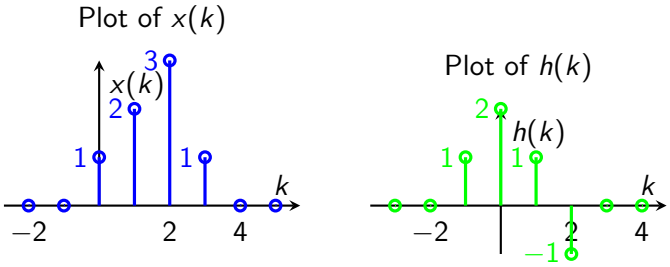
Example

The impulse response of a linear time-invariant system is

$$h(n) = \{1, \underset{\uparrow}{2}, 1, -1\}$$

Determine the response of the system to the input signal

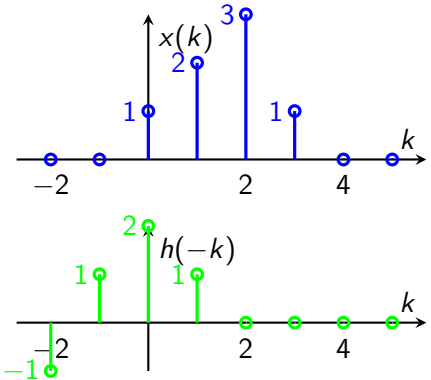
$$x(n) = \{\underset{\uparrow}{1}, 2, 3, 1\}$$



Example

Step 1. *Fold* (or time-reversal) $h(k)$ along the y-axis, to get $h(-k)$ as shown below. Then multiple and add each sample with the corresponding x(k). This will give us $y(0)$.
$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(-k)$.

$$y(0) = (0)(-1) + (0)(1) + (1)(2) + (2)(1) + (3)(0) + (1)(0) = 4$$

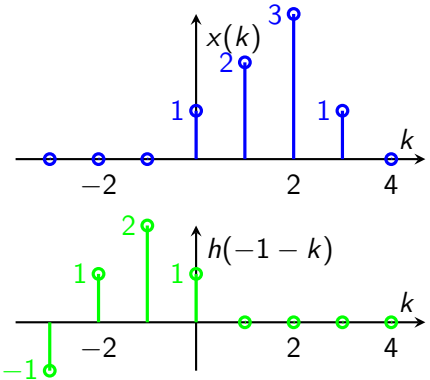


Example

Step 2. Shift and Multiply Then shift $h(-k)$ to the left by 1 to get $h(-1-k)$, and continue the operation. This will give us $y(-1)$. $y(-1) = \sum_{k=-\infty}^{\infty} x(k)h(-1-k)$.

$$y(-1) = (1)(1) = 1$$

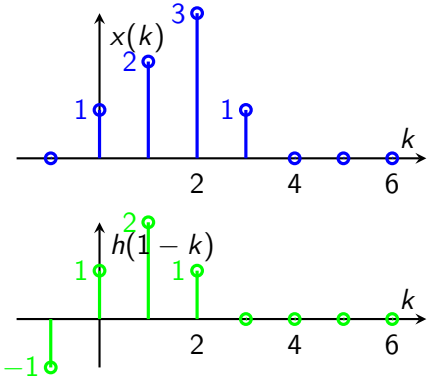
Further sifting to the left will result in $y(-2) = 0$, $y(-3) = 0$, etc



Example

Step 2. Shift and Multiply Then shift $h(-k)$ to the right by 1 give $h(1-k)$, multiply and add. This will give us $y(1)$. $y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1-k)$.

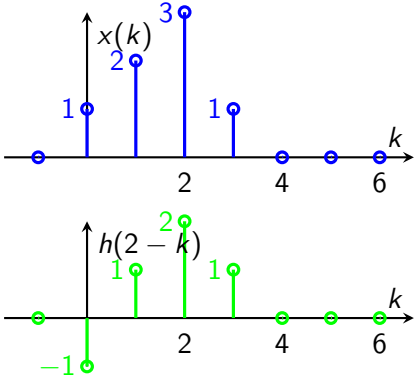
$$y(1) = (1)(1) + (2)(2) + (3)(1) = 81$$



Example

Step 2. Shift and Multiply Continue shifting $h(n-k)$ to the right by 1 which gives $h(2-k)$, $h(3-k)$, etc. Multiply and add. This will give us $y(2)$. $y(2) = \sum_{k=-\infty}^{\infty} x(k)h(2-k)$.

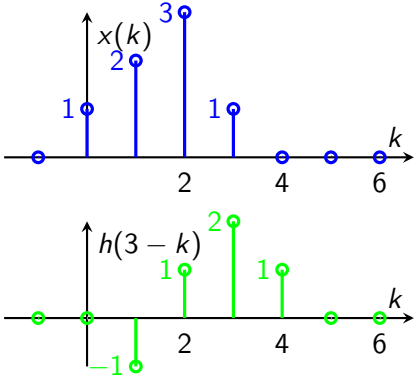
$$y(2) = (1)(-1) + (2)(1) + (3)(2) + (1)(1) = 8$$



Example

Step 2. Shift and Multiply Continue shifting $h(n-k)$ to the right by 1 which gives $h(3-k)$. Multiply and add. This will give us $y(3)$. $y(3) = \sum_{k=-\infty}^{\infty} x(k)h(3-k)$.

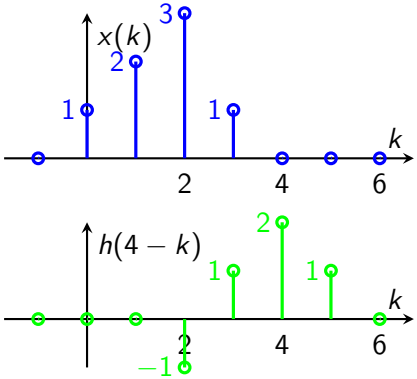
$$y(3) = (2)(-1) + (3)(1) + (1)(2) = 3$$



Example

Step 2. Shift and Multiply Shifting n for $h(4-k)$. This will give us $y(4)$.
$y(4) = \sum_{k=-\infty}^{\infty} x(k)h(4-k)$.

$$y(4) = (3)(-1) + (1)(1) = -2$$

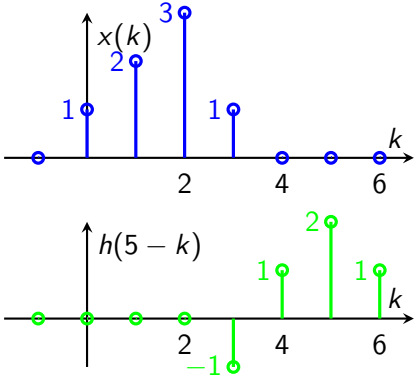


Example

Step 2. Shift and Multiply And lastly for $n = 5$ we have $h(5-k)$. This will give us $y(5)$. $y(5) = \sum_{k=-\infty}^{\infty} x(k)h(5-k)$.

$$y(5) = (1)(-1) = -1$$

Further shifting to the right will result in $y(6) = 0$, $y(7) = 0$, etc.

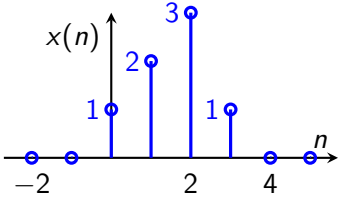


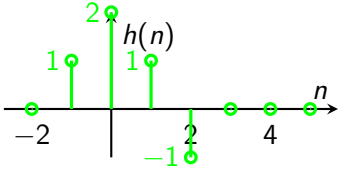
Example

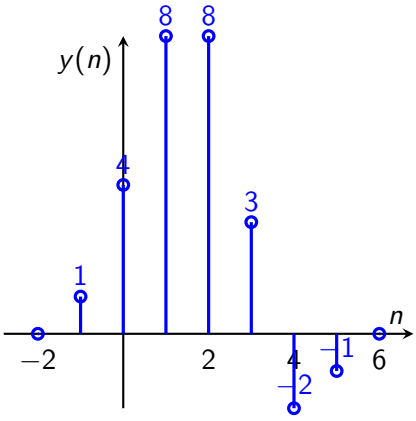
Step 3. Consolidate all the computed values of $y(n)$, We have

$$y(n) = \{1, \underset{\uparrow}{4}, 8, 8, 3, -2, -1\}$$

. A plot of the output is shown.







About Convolution

Option You can also choose to fold and shift $x(k) \rightarrow x(-k) \rightarrow x(n-k)$, instead of $h(k)$, this will give the same results since the convolution operation is commutative.

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(n-k)h(k)$$

Interpretation. In the above equation, For each sample of $y(n)$, is obtained as a linear combination of the impulse response $h(k)$, and the coefficients of the linear combination is determined by the values of $x(n-k)$.

Additional Properties of Discrete-time Convolution

Definition (Time-shift)

The **time-shift** property of the discrete time convolution states that if $h_1[k] * h_2[k] = c[k]$, then

$$h_1[k-m] * h_2[k-n] = c[k-m-n]$$

Definition (Convolution with Impulse)

Convolution with an **impulse** places the signal at the impulse location, i.e.

$$h[k] * \delta[k-m] = h[k-m]$$

Additional Properties of Discrete-time Convolution

Definition (Width of convolution)

The **width** property of the discrete time convolution states that if $h_1[k]$ has length m and $h_2[k]$ has length n, then $h_1[k] * h_2[k] = c[k]$ has length $m + n - 1$

In our previous example $x[n]$ has $length = 4$, and $h[n]$ also has $length = 4$, then $y[n]$ has $length = 4 + 4 - 1 = 7$.

Convolution: Practice Exercise

Let

$$x[n] = \alpha^n u[n], \text{ with } 0 < \alpha < 1$$

and unit step impulse response

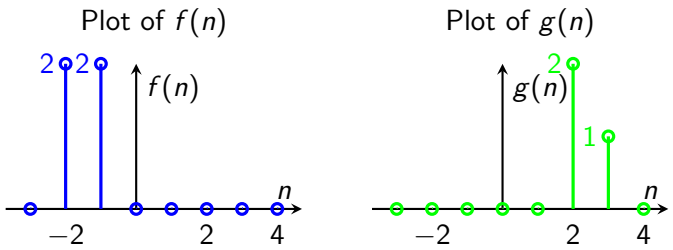
$$h[n] = u[n]$$

Show that

$$y[n] = x[n] * h[n] = \frac{1 - \alpha^{n+1}}{1 - \alpha} u[n], \text{ for } k \geq 0$$

Convolution for short signals

Consider the signals $f[n]$ and $g[n]$ as shown. Compute $h[n] = f[n] * g[n]$



Compute the starting point of n: $-2 + 2 = 0$
Compute the end point of n: $(-1) + 3 = 2$
This means that $h[n] = 0$ for $n < 0$ and $n > 2$
Only solve for $h[0]$, $h[1]$ and $h[2]$

Convolution for short signals

We need to evaluate

$$h[n] = \sum_{k=-\infty}^{\infty} f[k]g[n-k] = \sum_{k=-2}^{-1} f[k]g[n-k]$$

For $n = 0$:

$$h[0] = \sum_{k=-2}^{-1} f[k]g[0-k] = \sum_{k=-2}^{-1} f[k]g[-k]$$

$$h[0] = f[-2]g[2] + f[-1]g[1] = (2)(2) + (2)(0) = 4$$

For $n = 1$:

$$h[1] = \sum_{k=-2}^{-1} f[k]g[1-k] = f[-2]g[3] + f[-1]g[2] = (2)(1) + (2)(2) = 6$$

For $n = 2$:

$$h[2] = \sum_{k=-2}^{-1} f[k]g[2-k] = f[-2]g[4] + f[-1]g[3] = (2)(0) + (2)(1) = 2$$

Useful Table of Convolution Operations

Table: Useful convolution sums

$\mathbf{x_1[n]}$	$\mathbf{x_2[n]}$	$\mathbf{x_1[n] * x_2[n]}$
$\delta[n-k]$	$x[n]$	$x[n-k]$
$u[n]$	$u[n]$	$(n+1)u[n]$
$\alpha^n u[n]$	$u[n]$	$\left[\frac{1-\alpha^{n+1}}{1-\alpha}\right]u[n]$
$\alpha^n u[n]$	$\alpha^n u[n]$	$(n+1)\alpha^n u[n]$
$\alpha^n u[n]$	$\beta^n u[n]$	$\left[\frac{\alpha^{n+1}-\beta^{n+1}}{\alpha-\beta}\right]u[n]$

Iterative Solution

Given the 2nd-order difference equation:

$$y[n+2] - y[n+1] + 0.3y[n] = x[n+2] - 3x[n+1]$$

with the initial conditions $y[-1] = 1$ and $y[-2] = 2$ and $x[n] = nu[n]$. Rewriting the equation as:

$$y[n+2] = y[n+1] - 0.3y[n] + x[n+2] - 3x[n+1]$$

Case $n = -2$: $y[0] = y[-1] - 0.3y[-2] + x[0] - 3x[-1]$

$$y[0] = 1 - (0.3)(2) + 0 - 0 = 1 - 0.6 = 0.4$$

Case $n = -1$: $y[1] = y[0] - 0.3y[-1] + x[1] - 3x[0]$

$$y[1] = 0.4 - (0.3)(1) + 1 - 0 = 1.1$$

Continue solving the equation for all value of n iteratively.

Natural Response and Characteristic Polynomial

Natural Response: This is the response of the system when $x[n] = 0$, due to the initial conditions. (homogeneous solution). That is in the general form of a DT-LTI system, we have:

$$y(n) = \sum_{k=1}^{N} a_k y(n-k)$$

The solution to this equation is of the form λ^n. We have

$$\sum_{k=0}^{N} a_k \lambda^{n-k} = 0$$

Expanding

$$a_0\lambda^n + a_1\lambda^{n-1} + ... + a_{N-1}\lambda^{n-N+1} + a_N\lambda^{n-N} = 0$$

Factoring λ^{n-N} :

$$\lambda^{n-N} \underbrace{\left(a_0\lambda^N + a_1\lambda^{N-1} + ... + a_{N-1}\lambda + a_N\right)}_{\text{characteristic polynomial}} = 0$$

Zero-input Response

The **characteristic polynomial** is of the form

$$(a_0\lambda^N + a_1\lambda^{N-1} + ... + a_{N-1}\lambda + a_N)$$

In general, it has N roots $(\lambda_1, \lambda_2, ..., \lambda_N)$, which can be real and distinct, repeated roots or complex conjugates, since the coefficients $a_0, a_1, ..., a_N$ are real.

Zero-input Response: The zero-input response is a linear combination of $C_i\lambda^n$ (characteristic modes) and is the general solution to the equation.

$$y_{zi}(n) = C_1\lambda_1^n + C_2\lambda_2^n + ... + C_N\lambda_N^n$$

whee $C_1, C_2, ..., C_N$ are constants, and they are solved by applying the initial conditions of the system.

Solving for the Zero-Input response

Consider the DT-LTI system described by

$$y[n+2] - 0.8y[n+1] - 0.09y[n] = 3x[n+2]$$

with initial conditions $y[-1] = 0, y[-2] = \frac{10}{3}$.

The characteristic equation is

$$\lambda^2 - 0.8\lambda - 0.09 = 0$$

$$(\lambda - 0.9)(\lambda + 0.1) = 0$$

Therefore; The roots (eigenvalues) of the characteristic equation are $\lambda_1 = 0.9$ and $\lambda_2 = -0.1$

Solving for the Zero-Input response

The roots are real and distinct. The zero-input response can be written as:

$$y_{zi}(n) = C_1(0.9)^n + C_2(-0.1)^n$$

Solving for C_1 and C_2 using the initial conditions.

Let $n = -1$: $y[-1] = 0 = C_1(0.9)^{-1} + C_2(-0.1)^{-1}$

$$C_1\tfrac{10}{9} + C_2(10) = 0$$

Let $n = -2$: $y[-2] = \frac{10}{3} = C_1(0.9)^{-2} + C_2(-0.1)^{-2}$

$$C_1\tfrac{100}{81} + C_2(100) = \tfrac{10}{3}$$

Solving, $C_1 = \frac{27}{100}$ and $C_2 = \frac{3}{100}$, the **zero-input response** is

$$y_{zi}(n) = \tfrac{27}{100}(0.9)^n + \tfrac{3}{100}(-0.1)^n$$

Example: Zero-Input Response

For a DT-LTI system described by the difference equation, determine the zero-input response.

$$y[n+2] - 0.6y[n+1] - 0.16 = 5x[n+2]$$

given the initial conditions $y[-1] = 0$ and $y[-2] = \frac{25}{4}$.

The characteristic equation is

$$\lambda^2 - 0.6\lambda - 0.16 = (\lambda + 0.2)(\lambda + 0.8) = 0$$

The roots are $\lambda_1 = -0.2$ and $\lambda_2 = 0.8$
The zero-input response is:

$$y_{zi}[n] = C_1(-0.2)^n + C_2(0.8)^n$$

Example: Zero-Input Response

Solving for the constants.

Let $n = -1$: $0 = -5C_1 + \frac{5}{4}C_2$

Let $n = -2$: $\frac{25}{4} = 25C_1 + \frac{25}{16}C_2$

The constants are $C_1 = \frac{1}{5}$ and $C_2 = \frac{4}{5}$

The **zero-input reponse** is:

$$y_{zi}[n] = \frac{1}{5}(-0.2)^n + \frac{4}{5}(0.8)^n$$

Example: Repeated roots

Given the DT-LTI system described by:

$$y[n+2] + 6y[n+1] + 9y[n] = 2x[n+2] + 6x[n]$$

with initial conditions: $y[-1] = -\frac{2}{3}$ and $y[-2] = \frac{1}{9}$

The characteristic equation is $\lambda^2 + 6\lambda + 9 = 0$

$$(\lambda + 3)^2 = 0$$

The roots are: $\lambda_1 = -3$ and $\lambda_2 = -3$

Example: Repeated roots

The roots are repeated. The zero-input response is written as:

$$y_{zi}(n) = C_1(-3)^n + C_2 n(-3)^n$$

Solving for C_1 and C_2 from the initial conditions.

Let $n = -1$: $-\frac{2}{3} = C_1(-3)^{-1} + C_2 n(-3)^{-1}$

$$-\frac{2}{3} = (C_1 - C_2)(-\tfrac{1}{3})$$

Let $n = -2$: $\frac{1}{9} = C_1(-3)^{-2} + C_2 n(-3)^{-2}$

$$\tfrac{1}{9} = (C_1 - 2C_2)(\tfrac{1}{9})$$

Solving: $C_1 = 3$ and $C_2 = 1$, the zero-input response is

$$y_{zi}(n) = 3(-3)^n + n(-3)^n = (-3)^n(3 + n)$$

Example: Complex roots

Given the DT-LTI system described by:

$$y[n+2] - 1.3y[n+1] + 0.663y[n] = x[n+2]$$

with initial conditions $y[-1] = 2$ and $y[-2] = 1$.

The characteristic equation is $\lambda^2 - 1.3\lambda + 0.663 = 0$

$$(\lambda - 0.65 + j0.49)(\lambda - 0.65 - j0.49) = 0$$

The roots are $\lambda = 0.65 \pm j0.49$, in polar form $\lambda \cong 0.814e^{\pm j0.646}$

The form of the zero-input response is

$$y_{zi}(n) = C|\lambda|^n cos(\beta n + \theta)$$

where C and θ are unknown parameters that are solved using the initial conditions

Example: Complex roots

The zero-input response is

$$y_{zi}(n) = C|\lambda|^n cos(\beta n + \theta)$$

Solving the parameters using the initial conditions.

$$|\lambda| = |0.819e^{+j0.646j}| = |0.814| \underbrace{|e^{+j0.646}|}_{\text{magnitude}=1} = 0.814$$

$$\beta = \measuredangle 0.814e^{j0.646} = 0.646$$

So we have:

$$y_{zi}(n) = C|0.814|^n cos(0.646n + \theta)$$

Using the two initial conditions, solve for the two unknowns C and θ

Example: Complex roots

Given

$$y_{zi}(n) = C|0.814|^n cos(0.646n + \theta)$$

Using the two initial conditions, solve for the two unknowns C and θ.

Let $n = -1$: $2 = \frac{C}{0.814} cos(0.646(-1) + \theta)$

Let $n = -2$: $1 = \frac{C}{0.663} cos(0.646(-2) + \theta)$

Solving for C and θ (see details in the Appendix) we have:

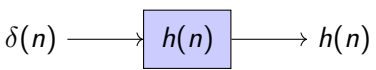
$$C = 1.942 \text{ and } \theta = 0.0696 \text{ radians}$$

The **zero-input response** is

$$y_{zi}(n) = 1.942|0.814|^n cos(0.646n + 0.0696)$$

Impulse response

The impulse response $h(n)$ of a DT-LTI system is the output response, with an impulse $\delta(n)$ as input and the system is initially at rest, or with **zero-initial states**. (zero initial conditions).



We can use the **iterative method** to solve for the first few samples of the **impulse response** $h(n)$

Impulse response

Determine the **impulse response** of the DT systems described by:

$$y[n] - 0.6y[n-1] - 0.16y[n-2] = 5x[n]$$

It follows that $x[n] = \delta[n]$ and $y[n] = h[n]$

$$h[n] - 0.6h[n-1] - 0.16h[n-2] = 5\delta[n]$$

Initial conditions: system is initially at rest, so

$$h[-1], h[-2], h[-3], etc = 0$$

We rewrite the equation for iterative solution as:

$$h[n] = 0.6h[n-1] + 0.16h[n-2] + 5\delta[n]$$

Impulse response: Iterative solution

At $n = 0$: $h[0] = 0.6\underbrace{h[-1]}_{0} + 0.16\underbrace{h[-2]}_{0} + 5\underbrace{\delta[0]}_{1} = 5$

At $n = 1$: $h[1] = 0.6\underbrace{h[0]}_{5} + 0.16\underbrace{h[-1]}_{0} + 5\underbrace{\delta[1]}_{0} = (0.6)(5) = 3$

At $n = 2$: $h[2] = 0.6h[1] + 0.16h[0] + 5\delta[2] = (0.6)(3) + (0.16)(5) = 2.6$

Continue solving for $h[3], h[4], h[5]$, etc. iteratively.

Being able to solve for a few values of $h(n)$ as initial conditions, can help solve for the closed-form solution of the impulse response $h(n)$

Impulse response: Closed-form solution

The impulse response of the system cam be computed from the zero-input response, with the initial conditions dictated by the impulse $\delta(n)$ at $n = 0$. That is for a causal $h(n)$

$$h(n) = \frac{b_0}{a_0}\delta(n) + y_{zi}(n)u(n)$$

where the coefficients a_0 and b_0 are the coefficients of the general-form of the difference equation, and $y_{zi}(n)$ is a linear combination of the characteristic modes of the system.

The n unknown coefficients in $y_{zi}(n)$ can be determined from knowledge of the first n values of $h(n)$. Fortunately, this can be computed iteratively as demonstrated earlier.

Impulse response: Closed-form Solution

Determine the impulse response $h(n)$ specified by the difference equation:

$$y[n] - 0.6y[n-1] - 0.16y[n-2] = 5x[n]$$

The equation can be advanced by setting $n = n + 2$

$$y[n+2] - 0.6y[n+1] - 0.16y[n] = 5x[n+2]$$

Therefore $a_0 = -0.16$ and $b_0 = 0$.
The characteristic equation is:

$$\lambda^2 - 0.6\lambda - 0.16 = (\lambda + 0.2)(\lambda - 0.8)$$

The characteristic modes are: $(-0.2)^n$ and $(0.8)^n$
The **zero-input response** is:

$$y_{zi}[n] = C_1(-0.2)^n + C_2(0.8)^n$$

Impulse response: Closed-form solution

The **impulse response** therefore is:

$$h[n] = (C_1(-0.2)^n + C_2(0.8)^n)u[n]$$

To determine the values of C_1 and C_2, we need two values of $h[n]$. This has been computed earlier with $h[0] = 5$ and $h[1] = 3$.

Using the initial conditions, we have $5 = C_1 + C_2$ and $3 = 0.2C_1 + 0.8C_2$. Therefore,

$$C_1 = 1 \text{ and } C_2 = 4$$

The **impulse response** is:

$$h[n] = [(-0.2)^n + 4(0.8)^n]u[n]$$

System response to external input: Zero-state response

The **zero-state** response $y[n]$ is the system response to an input $x[n]$, when the system is in zero-state, or zero-initial conditions.

The **zero-state response** $y_{zs}[n]$ of the system is the input $x[n]$ convolved with the impulse response $h[n]$.

$$y_{zs}[n] = x[n] * h[n]$$

Zero-state response

Calculate the **zero-state** response $y_{zs}[n]$ of a DT-LTI system described by:

$$y[n+2] - 0.6y[n+1] - 0.16y[n] = 5x[n+2]$$

given the input $x[n] = (4^{-n})u[n]$

In the previous example, the **impulse response** of this system is:

$$h[n] = [(-0.2)^n + 4(0.8)^n]u[n]$$

The zero-state response is $y_{zs}[n] = x[n] * h[n]$

$$y_{zs}[n] = (4^{-n})u[n] * [(-0.2)^n + 4(0.8)^n]u[n]$$

$$y_{zs}[n] = (4^{-n})u[n] * (-0.2)^n u[n] + (4^{-n})u[n] * 4(0.8)^n u[n]$$

Zero-state Response

Note that $(4^{-n})u[n] = (\frac{1}{4})^n u[n] = (0.25)^n u[n]$
Therefore

$$y_{zs}[n] = (0.25^n)u[n] * (-0.2)^n u[n] + 4(0.25^n u[n] * (0.8)^n u[n]$$

From the table of convolutional sum:

$$\begin{aligned} y_{zs}[n] &= \left[\frac{(0.25)^{n+1} - (-0.2)^{n+1}}{0.25 - (-0.2)} + 4\frac{(0.25)^{n+1} - (0.8)^{n+1}}{0.25 - 0.8}\right] u[n] \\ &= \left(2.22[(0.25)^{n+1} - (-0.2)^{n+1}] - 7.27[(0.25)^{n+1} - (0.8)^{n+1}]\right) u[n] \\ &= [-5.05(0.25)^{n+1} - 2.22(-0.2)^{n+1} + 7.27(0.8)^{n+1}]u[n] \end{aligned}$$

The **zero-state response** of the system is:

$$y_{zs}[n] = [-1.26(0.25)^n + 0.444(-0.2)^n + 5.81(0.8)^n]u[n]$$

Total Response

The **total response** of a DT-LTI system can be expressed as the sum of the zero-input response and the zero-state response:

$$y_{total}[n] = y_{zi}[n] + y_{zs}[n]$$

where

$$y_{total}[n] = \underbrace{\sum_{j=1}^{n} C_j \lambda_j^n}_{\text{zero-input}} + \underbrace{x[n] * h[n]}_{\text{zero-state}}$$

Example: Total Response

Determine the **total response** of the DT-LTI system defined by

$$y[n+2] - 0.16y[n+1] - 0.16y[n] = 5x[n+2]$$

with initial conditions $y[-1] = 0$, $y[-2] = \frac{25}{4}$ and input $x[n] = (4)^{-n}u[n]$ From the previous examples we have determined that the **impulse response** is given as:

$$h[n] = [(-0.2)^n + 4(0.8)^n]u[n]$$

The **zero-input response** as:

$$y_{zi}[n] = \frac{1}{5}(-0.2)^n + \frac{4}{5}(0.8)^n$$

And the **zero-state response** is:

$$y_{zs}[n] = [-1.26(0.25)^n + 0.444(-0.2)^n + 5.81(0.8)^n]u[n]$$

Example: Total Response

The total response therefore is given as:

$$y_t[n] = \underbrace{(0.2(-0.2)^n + 0.8(0.8)^n}_{\text{zero-input component}} \underbrace{-1.26(0.25)^n + 0.444(-0.2)^n + 5.81(0.8)^n)}_{\text{zero-state component}} u[n]$$

Note: The characteristic modes of this system are $(-0.2)^n$ and $(0.8)^n$. The zero-input component is made up of characteristic modes exclusively, but there are also characteristic modes in the zero-state response. When all the characteristic modes are lumped together, the resulting component is the **natural response** and the remaining part is the **forced response**. In this example:

$$y_t[n] = \underbrace{0.644(-0.2)^n + 6.61(0.8)^n}_{\text{Natural response}} \underbrace{-1.26(4)^{-n}}_{\text{Forced response}} u[n]$$

Natural and Forced Response

To solve for the natural and forced response of a system from the difference equation, then the classical method of solving for the homogeneous and particular solution is used similar to the one used to solve for differential equation in the continuous case. But we will not show it here. Refer to the appendix for an example.

Appendix: Complex roots complete solution

Solving for the unknown parameters C and θ given the two equations and applying the initial conditions.

Let $n = -1$: $2 = \frac{C}{0.814}\cos(0.646(-1) + \theta)$

Let $n = -2$: $1 = \frac{C}{0.663}\cos(0.646(-2) + \theta)$

Using the trigonometric identity $\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$

$$2 = \frac{C}{0.814}[\cos(0.646)\cos(\theta) + \sin(0.646)\sin(\theta)]$$

$$1 = \frac{C}{0.663}[\cos(1.292)\cos(\theta) + \sin(1.292)\sin(\theta)]$$

Appendix: Complex roots complete solution

Simplifying we have:

$$2 = C[0.981\cos(\theta) + 0.740\sin(\theta)] = 0.981C\cos(\theta) + 0.740C\sin(\theta)$$

$$1 = C[0.415\cos(\theta) + 1.450\sin(\theta)] = 0.415\underbrace{C\cos(\theta)}_{x} + 1.450\underbrace{C\sin(\theta)}_{y}$$

We have

$$0.981x + 0.740y = 2 \text{ and } 0.415x + 1.450y = 1$$

Solving for x and y: $x = 1.937 = C\cos(\theta)$ and $y = 0.135 = C\sin(\theta)$

$$\frac{C\sin(\theta)}{C\cos(\theta)} = \tan(\theta) = \frac{0.135}{1.937} \Rightarrow \theta = \tan^{-1}(\tfrac{0.135}{1.937})$$

Therefore: $\theta = 0.0696$ radians and $C = 1.942$

Appendix: Classical Method

We present the classical method of solving difference equations. The total solution $y_t[n]$ is the sum of the homogeneous solution and the particular solution.

$$y_t[n] = y_h[n] + y_p[n]$$

where $y_h[n]$: **homogeneous solution** is the solution when the input is zero. i.e. zero-input response
and $y_p[n]$: **particular solution** takes the form of the input.

Particular solution

The particular solution is assumed to take the form of the input, the table shows the general forms of $y_p[n]$ given $x[n]$

Table: General forms of $y_p[n]$

$x[n]$	$y_p[n]$
A (constant)	K (constant)
$A(m)^n$ (m is constant)	$K(M)^n$ (M is constant)
An^M	$K_0n^M + K_1n^{M-1} + ... + K_M$
$A^n n^M$	$A^n(K_0n^M + K_1n^{M-1} + ... + K_M)$
$Acos(\omega_0 n)/Asin(\omega_0 n)$	$K_1 cos(\omega_0 n) + K_2 sin(\omega_0 n)$

Example: Particular solution

Determine the particular solution of

$$y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n]$$

when $x[n] = 2^n, n \geq 2$ and 0 elsewhere

The form of the particular solution is $y_p[n] = K(2)^n, n \geq 0$ according to the table. Since its a solution it should satisfy the difference equation, substituting:

$$K(2)^n u[n] = \frac{5}{6}K2^{n-1}u[n-1] - \frac{1}{6}K2^{n-2}u[n-2] + 2^n u[n]$$

To solve for K, find a value of n where none of the terms vanish, for this case $n = 2$, since all terms exist for $n \geq 2$.

Particular solution

Since $u[2] = u[1] = u[0] = 1$, we have

$$4K = \frac{5}{6}K(2) - \frac{1}{6}K + 4$$

Solving for K, we have $K = \frac{8}{5}$

Therefore

$$y_p[n] = \frac{8}{5}(2)^n u[n]$$

Total solution

Solve for the **total solution** of the system described by

$$y[n] = -0.5y[n-1] + 0.5y[n-2] + x[n] - x[n-1]$$

with input $x[n] = (0.5)^n u[n]$, initial conditions $y[-1] = 1$, $y[-2] = -1$

Homogeneous solution $y_h[n]$:

$$y_h[n] + \tfrac{1}{2}y_h[n-1] - \tfrac{1}{2}y_h[n-2] = 0$$

The characteristic equations is:

$$\lambda^2 + \tfrac{1}{2}\lambda - \tfrac{1}{2} = (\lambda + 1)(\lambda - \tfrac{1}{2}) \rightarrow \lambda_1 = -1, \lambda_2 = \tfrac{1}{2}$$

The **homogeneous solution** is:

$$y_h[n] = C_1(-1)^n + C_2(\tfrac{1}{2})^n$$

Particular solution

Since the input is $x[n] = (0.5)^n u[n]$, has the form that is equal to one of the homogeneous solutions.
The form of the **particular solution** is:

$$y_p[n] = Kn(0.5)^n u[n]$$

Since $y_p[n]$ is a solution, we can substitute:

$$Kn(\tfrac{1}{2})^n u[n] = -\tfrac{1}{2}K(n-1)(\tfrac{1}{2})^{n-1}u[n-1] + \tfrac{1}{2}K(n-2)(\tfrac{1}{2})^{n-2}u[n-2] + (\tfrac{1}{2})^n u[n] - (\tfrac{1}{2})^{n-1}u[n-1]$$

Solving for K, at $n = 2$ none of the terms vanish, we have:

$$K2(\tfrac{1}{2})^2 = -\tfrac{1}{2}K(2-1)(\tfrac{1}{2})^{2-1} + (\tfrac{1}{2})K(2-2)(\tfrac{1}{2})^{2-2} + (\tfrac{1}{2})^2 - (\tfrac{1}{2})^{2-1}$$

$$\tfrac{1}{2}K = -\tfrac{1}{4}K + \tfrac{1}{4} - \tfrac{1}{2} \rightarrow K = -\tfrac{1}{3}$$

Total solution

The total solution therefore is:

$$y_t[n] = [C_1(-1)^n + C_2(\tfrac{1}{2})^n - \tfrac{1}{3}n(\tfrac{1}{2})^n]u[n]$$

To solve for C_1 and C_2, we have to use $n \geq 0$, so that none of the terms vanish. We cannot use $y[-1]$ and $y[-2]$.

We can solve for $y[0]$ and $y[1]$ by iteration from the original difference equation:

$$y[n] = -0.5y[n-1] + 0.5y[n-2] + x[n] - x[n-1]$$

with input $x[n] = (0.5)^n u[n]$, initial conditions $y[-1] = 1$, $y[-2] = -1$

Total solution

$$\text{At } n = 0 : y[0] = -\tfrac{1}{2}y[-1] + \tfrac{1}{2}y[-2] + x[0] - x[-1]$$
$$= -\tfrac{1}{2}(1) + \tfrac{1}{2}(-1) + 1 = 0$$

$$\text{At } n = 1 : y[1] = -\tfrac{1}{2}y[0] + \tfrac{1}{2}y[-1] + x[1] - x[0]$$
$$= -\tfrac{1}{2}(0) + \tfrac{1}{2}(1) + \tfrac{1}{2} - 1 = 0$$

Using $y[0] = 0$ and $y[1] = 0$ as initial conditions, and using the expression for $y_t[n]$ we have:

$$y[0] = C_1 + C_2 = 0$$

$$y[1] = -C_1 + \tfrac{1}{2}C_2 - \tfrac{1}{6} = 0$$

Solving we have: $C_1 = -\frac{1}{9}$ and $C_2 = \frac{1}{9}$, the **total solution** is

$$y_t[n] = [-\tfrac{1}{9}(-1)^n + \tfrac{1}{9}(\tfrac{1}{2})^n - \tfrac{1}{3}n(\tfrac{1}{2})^n]u[n]$$

Random Signals

- Are adjacent samples dependent?

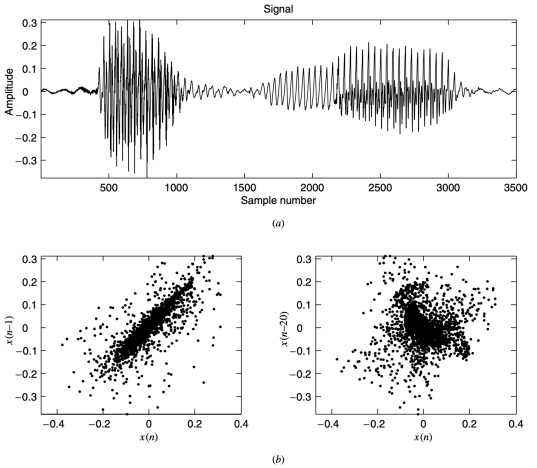


Figure: Speech signal